

SES4U ISP – Becoming the Expert

Overview

As up-and-coming scientific journalists, you are tasked with the important role of disseminating scientific knowledge to the public. You will delve into a topic relevant to the SES4U course, specializing in an aspect that piques your interest, and share this knowledge in a seminar presentation. Remember, you are informers, educators, translators, storytellers, and watchdogs. Your work will contribute to the public's understanding, thereby aiding in informed decision-making.

Objectives

The goal is to thoroughly research and present information on a chosen topic that aligns with the themes of Earth, Solar System, or Universe. Your 5-minute seminar should be enlightening and engaging to your classmates and guests.

Topic Selection

Choose a topic that connects to one of the main themes of the course. Here are some sample topics to inspire you:

- Astrobiology
- Humans in Space
- Extended Space Missions
- Technology in Space
- The usefulness of Meteors, Comets, Asteroids
- Exoplanets and Earth-like Planets
- The International Space Station: Cost vs. Benefit
- Canadian Space Program
- SpaceX and Private Space Ventures
- Dangers in the Universe
- Space Debris
- NASA's Direction Post Shuttle Program
- Socioeconomic Impacts of Natural Disasters (Landslides, Earthquakes, Volcanoes)
- Fossil Formation and Discovery
- Spin-off Innovations from Space Programs

Feel free to be creative and explore topics that are related to Earth and Space that may not be listed above.

Seminar Requirements

Your seminar must include the following elements:

- **History:** Outline the origin, development, and key milestones of your topic.
- **Location:** Specify the geographic or spatial relevance, if applicable.
- **Description:** Provide a detailed explanation of your subject. Demonstrate your expertise.
- **Significance/Importance:** Explain why your topic is critical to understand.
- **Technology Used:** Highlight the technological aspects associated with your topic.
- **Environmental/Social & Economic Impacts:** Discuss the broader implications of your topic.
- **References:** Cite your sources in an accepted format.
- **Duration:** Your presentation should last approximately 5 minutes.

Special emphasis must be placed on technology, and the environmental/social/economic impacts. These are crucial to your seminar, and your ability to connect your topic to these areas is essential.

Presentation Tools

Diversify your presentation methods by incorporating more than one tool. While Prezi and PowerPoint are valuable, you are encouraged to think outside the box. Consider creating a website, video, model, or other innovative means to captivate your audience. Remember, your presentation should reflect your expertise, and you should be able to speak freely and answer questions without relying on written notes.

Project Timeline

- **Introduction to Project, Research:** April 30, 2026
- **Topic Selection Due:** Refer to the Google Calendar
- **Work Periods:** Refer to the Google Calendar
- **Teacher/Student Conference:** Refer to the Google Calendar
- **Final Work Period / Conference:** Refer to the Google Calendar
- **Seminars:** Refer to the Google Calendar

Please consult the Google Calendar for specific dates and ensure you adhere to the timeline for a successful project completion.

Independent Study Project Rubric

Criteria	Unacceptable (0-1)	Needs Improvement (2-3)	Satisfactory (4-5)	Good (6-7)	Excellent (8-10)
Presentation Clarity and Engagement (15%)	Presentation lacks clarity and fails to engage audience	Presentation is somewhat clear but fails to engage audience	Presentation is clear and somewhat engaging	Presentation is clear and engaging for most of the audience	Presentation is exceptionally clear and engaging for the entire audience
Content Depth and Source Variety (10%)	Content lacks depth and sources are not varied	Content shows some depth but sources are not varied	Content has depth and sources are somewhat varied	Content is quite deep and sources are varied	Content is extremely deep and sources are very diverse
Progress during Check-ins (5%)	No progress made during check-ins	Little progress made during check-ins	Some progress made during check-ins	Good progress made during check-ins	Excellent progress made during check-ins
Topic Coverage: History, Location, Description, Significance, and Technological Aspects (20%)	Coverage of the topic is incomplete or inaccurate	Coverage of the topic is somewhat complete but lacks accuracy	Topic is adequately covered with minor inaccuracies	Topic is well-covered and accurate	Topic is thoroughly covered and highly accurate
Understanding of Environmental, Social, and Economic Impacts (20%)	Understanding of impacts is poor or lacking	Understanding of impacts is superficial	Understanding of impacts is adequate	Understanding of impacts is good	Understanding of impacts is excellent and nuanced
Use of Diverse Presentation Tools (15%)	Does not use any diverse presentation tools	Uses diverse presentation tools but not effectively	Uses diverse presentation tools effectively	Uses a variety of diverse presentation tools effectively	Uses a wide variety of diverse presentation tools very effectively

Ability to Speak Freely and Handle Queries (15%)	Unable to speak freely or handle queries	Struggles to speak freely and handle queries	Able to speak freely and handle some queries	Able to speak freely and handle most queries	Able to speak freely and handle all queries with expertise
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